

## BRATOLYUBOVKA, Russian Federation

WMO: 315210

Lat: 50.7863N

Lon: 129.4270E

Elev: 755

StdP: 14.30

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -33.6      | -29.6 | -38.1                         | 0.6 | -33.4 | -34.5 | 0.8 | -29.2 | 12.0                  | -2.0 | 10.6 | -4.3 | 2.4                   | 110  | 0.549 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.3                   | 85.5            | 69.8 | 82.6 | 69.2 | 79.8 | 67.9 | 73.1                | 81.5 | 71.5 | 79.6 | 70.0 | 76.9 | 4.3                  | 250  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.5                            | 115.4 | 77.6 | 68.9 | 109.0 | 75.7 | 67.4 | 103.6 | 74.1 | 37.4          | 81.7 | 35.9 | 79.7 | 34.6 | 77.0 | 79.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 15.0          | 12.2 | 10.6 | -39.1                      | 89.8 | 4.8                | 2.9 | -42.5                                              | 91.9 | -45.3      | 93.6 | -48.0      | 95.3 | -51.4      | 97.4 |    |    |
| (5)               |      |      | -39.5                      | 75.4 | 4.7                | 2.1 | -42.9                                              | 76.9 | -45.6      | 78.1 | -48.2      | 79.2 | -51.6      | 80.7 |    |    |

## Monthly Climatic Design Conditions

|             |                                                                                  | Annual  | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|-------------|----------------------------------------------------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|             |                                                                                  | (d)     | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)         | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg   | 31.8  | -12.3 | -2.5  | 15.2  | 37.8  | 53.6  | 64.2  | 69.5  | 65.7  | 52.9  | 35.2  | 10.3  | -10.2 | (6)  |
| (7)         |                                                                                  | DBStd   | 30.50 | 8.94  | 9.78  | 11.50 | 9.42  | 7.23  | 6.46  | 4.34  | 5.42  | 8.20  | 9.15  | 12.58 | 9.76  | (7)  |
| (8)         |                                                                                  | HDD50   | 8493  | 1933  | 1470  | 1078  | 383   | 43    | 0     | 0     | 1     | 63    | 465   | 1191  | 1866  | (8)  |
| (9)         |                                                                                  | HDD65   | 12429 | 2398  | 1890  | 1543  | 815   | 361   | 91    | 11    | 54    | 369   | 925   | 1641  | 2331  | (9)  |
| (10)        |                                                                                  | CDD50   | 1844  | 0     | 0     | 0     | 18    | 153   | 427   | 603   | 488   | 150   | 5     | 0     | 0     | (10) |
| (11)        |                                                                                  | CDD65   | 305   | 0     | 0     | 0     | 1     | 6     | 68    | 149   | 76    | 5     | 0     | 0     | 0     | (11) |
| (12)        |                                                                                  | CDH74   | 2510  | 0     | 0     | 0     | 16    | 143   | 737   | 1000  | 555   | 59    | 0     | 0     | 0     | (12) |
| (13)        |                                                                                  | CDH80   | 580   | 0     | 0     | 0     | 1     | 27    | 233   | 229   | 85    | 5     | 0     | 0     | 0     | (13) |
| (14)        | Wind                                                                             | WSAvg   | 4.3   | 3.4   | 3.8   | 4.7   | 5.7   | 5.6   | 4.0   | 3.8   | 3.9   | 4.4   | 5.0   | 4.3   | 3.5   | (14) |
| (15)        | Precipitation                                                                    | PrecAvg | 21.8  | 0.3   | 0.3   | 0.4   | 0.9   | 2.0   | 3.2   | 4.8   | 4.3   | 2.6   | 1.3   | 0.7   | 0.5   | (15) |
| (16)        |                                                                                  | PrecMax | 31.3  | 0.8   | 1.1   | 1.3   | 2.2   | 5.2   | 8.4   | 9.8   | 9.1   | 5.7   | 2.8   | 1.6   | 1.3   | (16) |
| (17)        |                                                                                  | PrecMin | 15.2  | 0.1   | 0.0   | 0.0   | 0.0   | 0.4   | 1.0   | 1.0   | 1.7   | 0.5   | 0.4   | 0.2   | 0.1   | (17) |
| (18)        |                                                                                  | PrecStd | 4.0   | 0.2   | 0.2   | 0.4   | 0.6   | 1.2   | 1.7   | 2.3   | 1.8   | 1.4   | 0.7   | 0.4   | 0.3   | (18) |
| (19)        | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%    | DB    | 14.3  | 28.3  | 46.8  | 76.1  | 84.1  | 91.4  | 89.0  | 85.3  | 80.1  | 65.7  | 44.6  | 22.6  | (19) |
| MCWB        |                                                                                  |         | 12.1  | 24.1  | 36.5  | 53.5  | 60.9  | 73.7  | 71.8  | 71.1  | 65.0  | 51.8  | 37.6  | 20.0  | (20)  |      |
| 2%          |                                                                                  | DB      | 9.3   | 23.8  | 41.9  | 65.9  | 78.2  | 86.1  | 85.3  | 82.4  | 75.2  | 58.2  | 38.5  | 14.3  | (21)  |      |
|             |                                                                                  | MCWB    | 7.7   | 20.6  | 33.2  | 48.8  | 57.6  | 67.9  | 72.0  | 70.7  | 62.9  | 46.6  | 33.0  | 13.1  | (22)  |      |
| 5%          |                                                                                  | DB      | 4.5   | 18.4  | 37.8  | 59.0  | 73.3  | 82.2  | 82.3  | 79.9  | 71.6  | 53.8  | 33.1  | 8.9   | (23)  |      |
|             |                                                                                  | MCWB    | 3.2   | 15.5  | 30.8  | 44.3  | 55.4  | 67.5  | 70.9  | 69.4  | 60.7  | 44.1  | 29.6  | 7.8   | (24)  |      |
| 10%         |                                                                                  | DB      | 0.9   | 13.0  | 33.2  | 53.9  | 69.2  | 78.0  | 79.5  | 77.1  | 67.4  | 49.5  | 29.4  | 4.9   | (25)  |      |
|             |                                                                                  | MCWB    | -0.1  | 10.9  | 28.1  | 41.2  | 53.8  | 65.0  | 69.7  | 67.4  | 58.7  | 41.6  | 26.5  | 3.8   | (26)  |      |
| (27)        | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%    | WB    | 12.8  | 24.3  | 37.2  | 54.6  | 63.4  | 75.6  | 75.3  | 75.0  | 67.3  | 53.0  | 38.4  | 20.5  | (27) |
| MCD B       |                                                                                  |         | 14.3  | 26.9  | 45.8  | 74.6  | 76.8  | 89.0  | 84.0  | 82.0  | 75.6  | 63.8  | 43.9  | 22.7  | (28)  |      |
| 2%          |                                                                                  | WB      | 7.8   | 20.4  | 33.7  | 49.1  | 59.8  | 71.1  | 73.3  | 72.5  | 64.8  | 49.2  | 33.7  | 13.3  | (29)  |      |
|             |                                                                                  | MCD B   | 9.2   | 23.9  | 40.4  | 63.8  | 73.8  | 81.9  | 81.6  | 79.7  | 72.2  | 56.1  | 38.3  | 14.3  | (30)  |      |
| 5%          |                                                                                  | WB      | 3.3   | 15.6  | 31.1  | 45.5  | 57.4  | 68.3  | 72.0  | 70.6  | 62.7  | 45.2  | 30.1  | 7.8   | (31)  |      |
|             |                                                                                  | MCD B   | 4.5   | 18.4  | 37.1  | 56.9  | 69.8  | 78.7  | 80.2  | 77.2  | 69.5  | 52.1  | 33.4  | 8.8   | (32)  |      |
| 10%         |                                                                                  | WB      | -0.1  | 11.0  | 28.4  | 42.3  | 55.4  | 66.4  | 70.8  | 68.8  | 60.2  | 42.2  | 26.2  | 3.9   | (33)  |      |
|             |                                                                                  | MCD B   | 0.8   | 13.1  | 33.2  | 52.9  | 66.2  | 75.6  | 78.2  | 74.6  | 65.9  | 48.8  | 29.2  | 4.9   | (34)  |      |
| (35)        | Mean Daily<br>Temperature<br>Range                                               | 5% DB   | MDBR  | 18.5  | 22.0  | 21.5  | 20.8  | 22.2  | 20.1  | 17.3  | 18.5  | 21.1  | 18.9  | 17.4  | 17.1  | (35) |
| MCDBR       |                                                                                  |         | 18.6  | 22.6  | 21.5  | 27.9  | 29.5  | 26.8  | 22.1  | 22.8  | 25.7  | 26.2  | 17.8  | 15.9  | (36)  |      |
| MCWBR       |                                                                                  |         | 17.3  | 19.3  | 14.3  | 15.4  | 14.0  | 12.0  | 10.5  | 11.8  | 13.8  | 16.1  | 13.7  | 14.8  | (37)  |      |
| 5% WB       |                                                                                  | MCDBR   | 18.5  | 22.4  | 19.8  | 26.1  | 25.5  | 23.2  | 18.4  | 18.4  | 21.4  | 22.4  | 17.4  | 15.7  | (38)  |      |
|             |                                                                                  | MCWBR   | 17.3  | 19.2  | 13.5  | 14.6  | 13.5  | 11.7  | 9.5   | 10.2  | 12.1  | 14.7  | 13.8  | 14.7  | (39)  |      |
| (40)        | Clear-Sky<br>Solar<br>Irradiance                                                 | taub    | 0.267 | 0.300 | 0.380 | 0.544 | 0.601 | 0.494 | 0.477 | 0.426 | 0.401 | 0.417 | 0.332 | 0.272 | (40)  |      |
| taud        |                                                                                  | 2.138   | 2.058 | 1.884 | 1.682 | 1.654 | 2.012 | 2.112 | 2.246 | 2.223 | 1.996 | 2.073 | 2.090 | (41)  |       |      |
| Ebn at Noon |                                                                                  | 236     | 257   | 251   | 220   | 215   | 245   | 247   | 255   | 247   | 206   | 200   | 209   | (42)  |       |      |
| Edh at Noon |                                                                                  | 24      | 35    | 50    | 69    | 75    | 53    | 47    | 39    | 36    | 37    | 26    | 22    | (43)  |       |      |
| (44)        | All-Sky Solar<br>Radiation                                                       | RadAvg  | 518   | 907   | 1407  | 1599  | 1727  | 1887  | 1750  | 1528  | 1205  | 787   | 508   | 401   | (44)  |      |
| (45)        |                                                                                  | RadStd  | 14    | 20    | 81    | 170   | 136   | 224   | 129   | 99    | 106   | 40    | 27    | 22    | (45)  |      |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (14 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page